AirTouch - Intelligent Virtual Multi-touch Screen

Research Proposal · August 2017

1 author:

Naveen Kumarasinghe
Sri Lanka Institute of Information Technology

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D. N. D. Kumarasinghe
Faculty of Computing
Sri Lanka Institute of Information Technology, Malabe, Sri Lanka
dndkumarasinghe@gmail.com

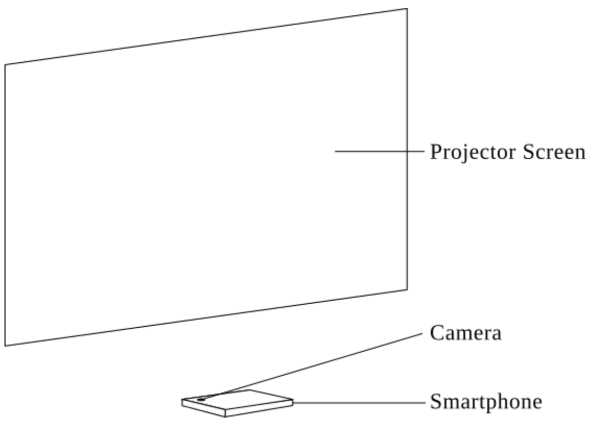
Abstract

This research elaborates a technology that allows smartphones to generate a touch-sensitive region in three-dimensional space. This allows us to keep a smartphone under a projector screen and transform the projector screen into an interactive virtual touch screen. When the projector screen is touched by an operator, the scene is observed by the phone's camera. Then the scene is sent to an image processing server to estimate the location of the touched points. These measurements are then enhanced by a regression-based machine learning model. Having known the touched coordinates, we can trigger the actual touch gestures at the operating system level. This system is a collaboration of a smartphone, a personal computer and a physical screen. Ultimately, this virtual touch screen aims to become a low-cost human interface device for computer systems that can be used to enhance teaching, presenting and gaming experiences.

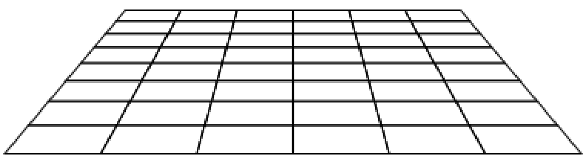
Keywords: human computer interaction, virtual touch screen, gesture recognition, computer interfaces, monocular computer vision

1 Introduction

Initially, the smartphone is placed below a projector screen as its camera faces up.



The position of the smartphone in three dimensional space. Due to this arrangement, the smartphone receives a distorted view of the projector screen.



The distorted projector screen plane is seen through the smartphone's camera. Both the distorted projector screen and the actual computer screen are composed of a finite amount of elements known as pixels.

Even though both screens display the same content the distorted screen accommodates a different amount of pixels from the actual screen depending on the camera's orientation and resolution. However the actual screen consists of a fixed amount of pixels defined by the operating system. In this study, we discuss the methods of mapping pixels on the distorted screen to the actual screen. When the mapping process is complete and a fingertip is placed on the projector screen, we can obtain the actual touchpoint on the computer screen. Then we trigger a touch gesture at the operating system level. It is also important to use a camera sensor that has a higher resolution than the computer screen for better accuracy.

2 Methodology

2.1 Preprocessing

First, it is essential to isolate the operator's hand from the scene as it provides key factors for calculating the touched coordinates. Assuming that there are no moving objects in the scene (ie. ceiling fans) we obtain a background image BG excluding the operator. When the operator interacts with the projector screen we obtain the foreground image FG which includes the operator's hand. Then a blur filter of a 5x5 kernel is applied to BG and FG to reduce the noise caused by the camera lens. As BG and FG are in form of matrices, we perform the following matrix operations to isolate the hand into image B, under indoor lighting conditions:

$$B = (BG - FG) * 4$$

B is then converted into a gray-scale image and thresholded at the intensity level of 100 to obtain the binary image. The largest contour in the resulting image can be considered as the operator's hand.

2.2 Calibration

Calibration is performed to establish a relation between two screens S and S_d. Each screen is two dimensional. The actual screen's dimensions X and Y, should be mapped to their respective dimensions X_d and Y_d on the distorted screen. Once the mapping process is complete, touchpoints on the distorted screen can be converted to the touchpoints on the actual screen.

2.2.1 Mapping Y-dimensions

Y on the actual screen in a linear dimension. However, Y_d on the distorted screen becomes a non-linear dimension due to the position of the camera. In order to map these two dimensions we use a solid horizontal scan line. The scan line is drawn on a white background, and is translated from bottom to top on the actual screen at a constant speed. The translation is also visible on the distorted screen.

Let the scan line on the actual screen be L and the scan line on the distorted screen be L_d. L is the independent variable, while L_d is the dependent variable. As we elevate L on the actual screen we record the position of L_d against the position of L. An algorithm was used to obtain L_d's position which performs a simple background subtraction and finds the longest horizontal line on the distorted screen. Once the recording process is complete we have a data set that reveals the relation between L and L_d

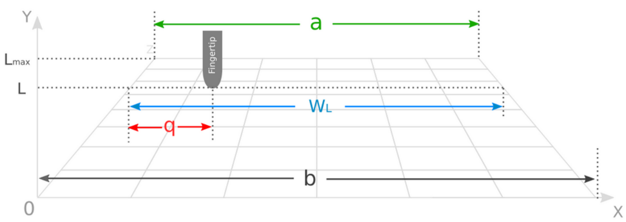
The data set is then smoothened by running through a cosine interpolation algorithm

Given the y coordinate of a fingertip on the distorted screen, now we can obtain the analogous Y coordinate on the actual screen using the relation.

As Y_d is typically a smaller dimension than Y, units on Y_d should be mapped to Y on one-to-many basis. This will cause spatial discrimination on the virtual touch screen which leads to a loss of precision.

2.2.1.1 Estimating the X-coordinate

Estimating X-coordinate on the computer screen is rather straightforward. Consider the below image:



a – Relative width of the top edge of the projector screen.

b – Relative width of the bottom edge of the projector screen.

W_L – Relative width of the projector screen at the given y-coordinate L.

q – Relative distance to the fingertip from the left edge of the projector.

L_{max} – Projector screen's top edge level relative to Y dimension.

L – Level of the fingertip relative to the Y dimension.

The boundary data a, b, Y_{max} are already learned by the system in the calibration phase. Note that when the level L increases, W_L (the relative width of the projector screen at L)decreases. L and W_L has an inverse linear relationship and can be written as a function:

$$W_L = m \, . \, L + b$$

where the gradient,

$$m = \frac{a-b}{L_{max}}$$

Then the system can obtain the x-coordinate of the computer screen from the following function:

$$g(q) = q \, . \, \frac{W_c}{W_L}$$

where,

W_c – Width of the computer screen.

The final CPU optimized function for obtaining X-coordinate can be written as:

$$g(q) = q \, . \, \frac{W_c}{L \frac{(a-b)}{L_{max}} + b}$$

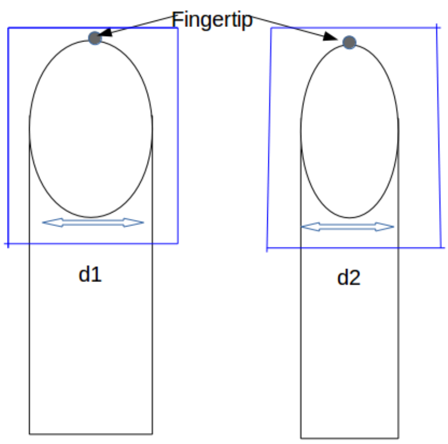
2.3 Gesture recognition

Gestures are substantial activities made by people to pass on important data to others. They are also signals that computers can percept. A gesture can be narrowed down into two particular classes; Dynamic and static.

A dynamic gesture is expected to change over a time frame though a static signal is seen at the spurt of time. A waving hand implies farewell is a case of element motion and the stop sign is a case of static motion. In this paper we primarily concentrate on close by gestures. The objective is to make a framework which can recognize particular hand signals and utilize them to pass on data to other systems.

2.3.1 Gesture Recognition Algorithm

Through this research we proposed a novel technique of hand gesture recognition and fingertip tracking based on the detection of some shape based features. Then we came up with an algorithm called "Tipy". Tipy has the capability to track 10fingertips. First this algorithm separates the largest contour which is probably the hand. Then we scan through the image to find white pixels and we scan the consecutive white pixels. When the number of white pixels exceed a certain threshold the algorithm counts the contour as a fingertip. Going back to the initial white pixel the algorithm travels to the topmost pixel where Y value is minimum. The starting point and the ending point of the white pixel are stored in the range table. The rest of the fingertips are scanning in the same manner. Before the algorithm attempts to scan the previously scanned fingertip again it checks entries in the range table if the new existing range is a superset of any previous range the algorithm will ignore the fingertip and add the new range to the new range table. This way we can get all the points of the fingertips.

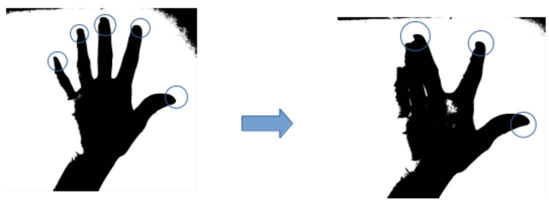


2.3.2 Gestures

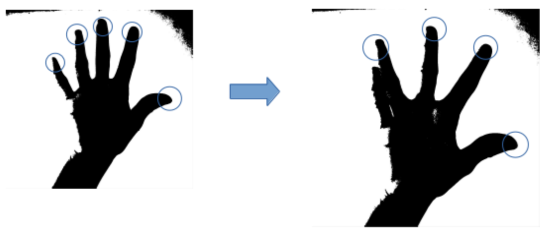
After tracking and recognizing fingertips through Tipy algorithm, the second task is to recognize different gestures and perform an action for each and every gesture. According to the research we must keep our hand above the camera and it is not essential to touch the projector screen to perform primary gestures. When the hand starts moving, movement of the hand is replicated by the cursor. The size of the virtual touch screen didn't have an adversed impact on gesture recognition. When the user move his hand the cursor moves along with it.

In this research area we mapped mouse pointer and the hand to perform gestures. Through this when the user holds his hand above the camera, camera will be sending the frames to the system and the system will recognize the number of folded fingers of the hand and identify it as a gesture. When the user combine fingers, the combination acts as a unique gesture. For example, if the system recognizes hand gesture with 4 fingertips It will be a left mouse click event. If it is a gesture with 3 fingertips it will be recognized as right mouse click event. Likewise we were able to map different gestures to unfolded finger tip patterns.

swipe-right/next



swipe-left/previous



Later we calculate the convex hull of the hand contour, The convex hull will wrap the convexes of the contour .

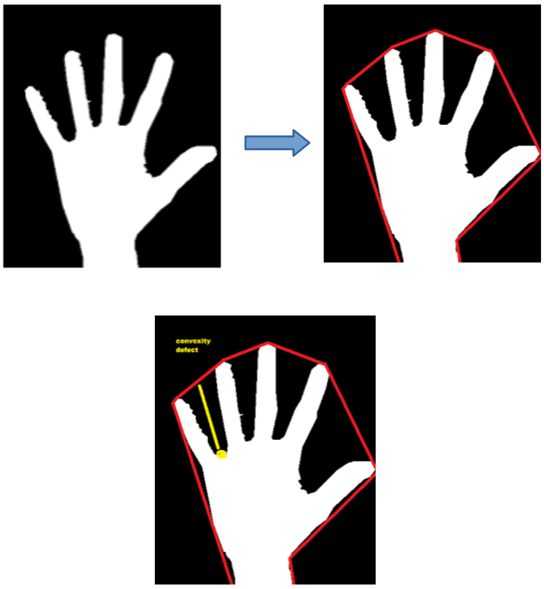


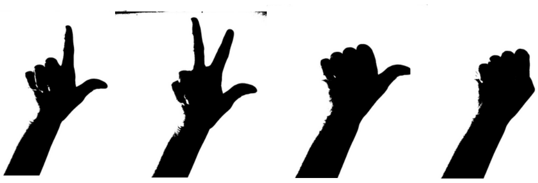
Figure – Convex Hull Mechnism

As of now we have the points of the convex hull as well as points of the fingertips given by the Tipy algorithm. When the fingers are unfolded the fingertips lie on the boarder of the convex hull, but if we fold one finger the tip of that fingertip falls below the boarder of the convex hull. So the distance to the convex hull from the fingertip will be larger than the others. So we can accurately calculate how many fingertips have been folded in a one hand.

Then we keep track of all the 5 fingers of the hand now we can monitor which fingers have been folded. Based on the folded fingertips we define gestures as bellow.

Also through the system we recognize zoom in and zoom out gestures to zoom pictures or relevant area. User must use both hands to perform zoom gestures. Two fingertips will perform zoom in action and one fingertip will perform zoom out gesture.

Recognized Gestures:



Mapping physical screen to touch screen.

The Laptop or computer screen pixel or ratio is different from projector screen. Every pixel in computer map into the projector screen. To do mapping the projector screen into virtual touch screen we are using simple machine learning techniques.

The system will send the scan line to the screen pixel by pixel and the camera will observes the scan line frame by frame. Then the system will remove the background and process the scan line using threshold algorithms based on that X and Y coordinates will map. All the pixel will map by machine learning techniques.

X and Y axis Scan line

Mapping Result

Laptop Screen X axis	Projector screen X axis
X=0 PX	X=6 PX
X=75 PX	X=126 PX
X=236 PX	X=365 PX
X=500 PX	X=756 PX

Laptop Screen Y axis	Projector screen Y axis
Y=0 PX	Y=0 PX
Y=75 PX	Y=250 PX
Y=236 PX	Y=856 PX
Y=1000 PX	Y=2150 PX

3 Conclusion

A method was developed to utilize smartphones to detect gestures and interface with computer systems. The accuracy of virtual touch screen significantly relied on hardware components such as the camera sensor and constant lighting conditions. Given the perfect test conditions, we noticed the accuracy is inversely propotional to the screen size. However the system had an acceptable respose to touch gestures with an error margin of 15%-20%. As we do not use specialized hardware, AirTouch as a virtual game controller was identified as a better application.

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